A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-26136466})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary divisor I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \mathrm{div}(t) + i_x(J) = 0, \qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 1835 & 997 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,1835 + \mathbf{Z}\,(997 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K. The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 + 26136466 = 0$, of discriminant -104545864. Class group invariants $(60 \quad 10 \quad 5)$, class number 3000; the three stored generator ideals (HNF) are $\begin{pmatrix} 7 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 5110 & 288 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1538 & 1082 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 6093y^3 - 28sy^2 + 840932y + 2672s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 12186*y^8 + 38806513*y^6 + 10243391992*y^4 - 3203685051888*y^
2 + 186603494470144
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{144237523}{20805604708146875} + (\frac{209}{20805604708146875})\, s, \qquad J = \begin{pmatrix} 1835 & 997 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1835$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2346748210944 = 2^8 \cdot 3^2 \cdot 163 \cdot 6248797$:

$$\begin{pmatrix} 12222646932 & 11519520870 & 4737325894 & 6286921160 & 5190493868 & 10955947053 & 10427831658 & 9817023442 & 3677316565 & 5326909892 \\ 0 & 6 & 2 & 0 & 0 & 2 & 4 & 4 & 2 & 4 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 403 factors with signed exponents of absolute value up to 228053440845485870640; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 1 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 6093y^3 - 28sy^2 + 840932y + 2672s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 12186*y^8 + 38806513*y^6 + 10243391992*y^4 - 3203685051888*y^
2 + 186603494470144
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{12837823}{2638790623394114} + \left(-\frac{265}{10555162493576456}\right) s, \qquad J = \begin{pmatrix} 2114 & 1252 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9461046529216 = 2^6 \cdot 53 \cdot 103 \cdot 139 \cdot 194819$:

$$\begin{pmatrix}
591315408076 & 265660819632 & 521711836282 & 42802742505 & 151118098892 & 559501203342 & 417390893879 & 148028157570 & 520310520659 & 78272553440 \\
0 & 2 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 403 factors with signed exponents of absolute value up to 70858621456585792934; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 6093y^3 - 28sy^2 + 840932y + 2672s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 12186*y^8 + 38806513*y^6 + 10243391992*y^4 - 3203685051888*y^
2 + 186603494470144
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{5708627}{537850646109698} + \left(\frac{793}{2151402584438792}\right) s, \qquad J = \begin{pmatrix} 1538 & 1082 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1538$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $60665404266752 = 2^8 \cdot 53 \cdot 4471211989$:

$$\begin{pmatrix} 947896941668 & 803744554748 & 581557369948 & 946718864699 & 678689935669 & 380723959298 & 367805455490 & 675716128712 & 332960481410 & 700551049686 \\ 0 & 4 & 0 & 2 & 3 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 403 factors with signed exponents of absolute value up to 828610105250069514479; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s + 435)\, y^3 + (27s - 179284)\, y^2 + (202s + 3993818)\, y + (-6930s - 12439896)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 874*y^8 - 360308*y^7 + 35030463*y^6 - 1608201308*y^5
+ 44161485350*y^4 - 795530052256*y^3 + 11696814898196*y^2 - 17254014
7715376*y + 1409952178494216
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{144237523}{20805604708146875} + \left(\frac{209}{20805604708146875}\right) s, \qquad J = \begin{pmatrix} 1835 & 997 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1835$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $55886436734328 = 2^3 \cdot 3^2 \cdot 619 \cdot 1253958821$:

$$\begin{pmatrix}
4657203061194 & 4639511317372 & 2374126973256 & 3467527206912 & 2716291005782 & 971578147824 & 1257640549771 & 509573532860 & 1467219880982 & 3955367584420 \\
0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 3 & 1 & 2 & 2 & 1 & 1 & 1 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 330 factors with signed exponents of absolute value up to 88093979730; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \quad 0 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s + 435)\, y^3 + (27s - 179284)\, y^2 + (202s + 3993818)\, y + (-6930s - 12439896)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 874*y^8 - 360308*y^7 + 35030463*y^6 - 1608201308*y^5
+ 44161485350*y^4 - 795530052256*y^3 + 11696814898196*y^2 - 17254014
7715376*y + 1409952178494216
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{12837823}{2638790623394114} + (-\frac{265}{10555162493576456})\, s, \qquad J = \begin{pmatrix} 2114 & 1252 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $90033655327182 = 2 \cdot 3^2 \cdot 7 \cdot 714552820057$:

$$\begin{pmatrix} 30011218442394 & 896715638588 & 7520965343796 & 12442824857520 & 26563304891072 & 24956227602771 & 5431377399388 & 27709369433721 & 12473296436582 & 104536855666 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 1 & 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 333 factors with signed exponents of absolute value up to 86314277059; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s + 435)\, y^3 + (27s - 179284)\, y^2 + (202s + 3993818)\, y + (-6930s - 12439896)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 874*y^8 - 360308*y^7 + 35030463*y^6 - 1608201308*y^5
+ 44161485350*y^4 - 795530052256*y^3 + 11696814898196*y^2 - 17254014
7715376*y + 1409952178494216
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{5708627}{537850646109698} + (\frac{793}{2151402584438792})\, s, \qquad J = \begin{pmatrix} 1538 & 1082 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1538$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1421394542708 = 2^2 \cdot 7^2 \cdot 7252012973$:

$$\begin{pmatrix} 101528181622 & 18304199278 & 75458192040 & 45029116622 & 70198358624 & 17452173720 & 67558878107 & 91857305288 & 16430314518 & 20342320430 \\ 0 & 2 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 7 & 1 & 5 & 6 & 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 331 factors with signed exponents of absolute value up to 117207437084; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 - 17y^3 + (-24s - 15189)\, y^2 + (48s - 2960434)\, y + (17616s + 62203704)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 33*y^8 - 30344*y^7 - 5890201*y^6 + 130844702*y^5 + 15
261557485*y^4 + 27598720452*y^3 - 15165395516780*y^2 - 3240996019296
96*y + 11980059248066112
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{144237523}{20805604708146875} + \left(\frac{209}{20805604708146875}\right) s, \qquad J = \begin{pmatrix} 1835 & 997 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1835$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $10616973765804 = 2^2 \cdot 3^2 \cdot 7^2 \cdot 67 \cdot 2273 \cdot 39521$:

$$\begin{pmatrix} 1769495627634 & 1329768656350 & 158446249034 & 1175013398748 & 999515143070 & 965840048054 & 1217710622649 & 1602131266300 & 1279525768812 & 1036935003991 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 6 & 4 & 4 & 2 & 3 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 293 factors with signed exponents of absolute value up to 4970391044; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 - 17y^3 + (-24s - 15189)\, y^2 + (48s - 2960434)\, y + (17616s + 62203704)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 33*y^8 - 30344*y^7 - 5890201*y^6 + 130844702*y^5 + 15
261557485*y^4 + 27598720452*y^3 - 15165395516780*y^2 - 3240996019296
96*y + 11980059248066112
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{12837823}{2638790623394114} + \left(-\frac{265}{10555162493576456}\right) s, \qquad J = \begin{pmatrix} 2114 & 1252 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $34336544468324 = 2^2 \cdot 7^2 \cdot 175186451369$:

$$\begin{pmatrix} 2452610319166 & 1780039898434 & 1388457492624 & 2286710704356 & 535664907641 & 1965598013425 & 1301819504696 & 486827927328 & 1478472405806 & 2045410874175 \\ 0 & 14 & 7 & 0 & 9 & 0 & 2 & 0 & 6 & 8 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 290 factors with signed exponents of absolute value up to 111925686846; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 - 17y^3 + (-24s - 15189)\, y^2 + (48s - 2960434)\, y + (17616s + 62203704)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 33*y^8 - 30344*y^7 - 5890201*y^6 + 130844702*y^5 + 15
261557485*y^4 + 27598720452*y^3 - 15165395516780*y^2 - 3240996019296
96*y + 11980059248066112
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{5708627}{537850646109698} + \left(\frac{793}{2151402584438792}\right) s, \qquad J = \begin{pmatrix} 1538 & 1082 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1538$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $10616973765804 = 2^2 \cdot 3^2 \cdot 7^2 \cdot 67 \cdot 2273 \cdot 39521$:

$$\begin{pmatrix}
1769495627634 & 1329768656350 & 158446249034 & 1175013398748 & 999515143070 & 965840048054 & 1217710622649 & 1602131266300 & 1279525768812 & 1036935003991 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 6 & 4 & 4 & 2 & 3 & 3 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 290 factors with signed exponents of absolute value up to 65467097980; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s - 1381)\, y^3 + (-40s + 73531)\, y^2 + (151s + 1127436)\, y + (11602s - 44474512)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 - 2753*y^8 + 155348*y^7 + 29857313*y^6 + 1792111018*y^5
+ 36484809669*y^4 - 633557471168*y^3 - 28932283470942*y^2 - 8707277
705800*y + 5496117919166408
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{144237523}{20805604708146875} + \left(\frac{209}{20805604708146875}\right) s, \qquad J = \begin{pmatrix} 1835 & 997 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1835$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $81512130868624 = 2^4 \cdot 2087 \cdot 23057 \cdot 105871$:

$$\begin{pmatrix}
20378032717156 & 696080944086 & 2756324354528 & 13880843975470 & 4947760541549 & 11777967861302 & 19951557836080 & 12262363762961 & 10818127748969 & 2137840979361 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 2 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 474 factors with signed exponents of absolute value up to 1598970961162; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 4 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a+b$, column e_2

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s - 1381)\, y^3 + (-40s + 73531)\, y^2 + (151s + 1127436)\, y + (11602s - 44474512)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 - 2753*y^8 + 155348*y^7 + 29857313*y^6 + 1792111018*y^5
+ 36484809669*y^4 - 633557471168*y^3 - 28932283470942*y^2 - 8707277
705800*y + 5496117919166408
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{12837823}{2638790623394114} + (-\frac{265}{10555162493576456})\, s, \qquad J = \begin{pmatrix} 2114 & 1252 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $21470527694016 = 2^6 \cdot 3^4 \cdot 4141691299$:

$$\begin{pmatrix}
49700295588 & 11341174740 & 33197422236 & 32148654198 & 4740276406 & 43695304878 & 4763190692 & 32110319916 & 18464477219 & 35458116294 \\
0 & 6 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 6 & 0 & 4 & 2 & 5 & 1 & 2 & 5 \\
0 & 0 & 0 & 6 & 2 & 0 & 0 & 3 & 3 & 3 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 459 factors with signed exponents of absolute value up to 856926091686; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 4 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a+b$, column e_3

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s - 1381)\, y^3 + (-40s + 73531)\, y^2 + (151s + 1127436)\, y + (11602s - 44474512)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 - 2753*y^8 + 155348*y^7 + 29857313*y^6 + 1792111018*y^5
+ 36484809669*y^4 - 633557471168*y^3 - 28932283470942*y^2 - 8707277
705800*y + 5496117919166408
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{5708627}{537850646109698} + (\frac{793}{2151402584438792})\, s, \qquad J = \begin{pmatrix} 1538 & 1082 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1538$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $372723413381824 = 2^6 \cdot 1609 \cdot 15359 \cdot 235661$:

$$\begin{pmatrix}
11647606668182 & 700271470046 & 9701728471874 & 5039156873040 & 559522516898 & 6574857273945 & 5572621467726 & 843398288634 & 4179136598726 & 3914810527097 \\
0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\
0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 460 factors with signed exponents of absolute value up to 1104732399212; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 4 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 811y^3 - 10sy^2 + 2139000y + 18000s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 1622*y^8 + 4935721*y^6 - 855811400*y^4 - 4833806760000*y^2 +
8468214984000000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{144237523}{20805604708146875} + (\frac{209}{20805604708146875})\, s, \qquad J = \begin{pmatrix} 1835 & 997 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1835$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $163504363514560 = 2^6 \cdot 5 \cdot 71 \cdot 7196494873$:

$$\begin{pmatrix}
10219022719660 & 5226585942720 & 8298689620642 & 3049750371276 & 3025955700618 & 6337798042082 & 1673122811974 & 6318472220382 & 9367974016070 & 7453893073132 \\
0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 271 factors with signed exponents of absolute value up to 2268879515; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 811y^3 - 10sy^2 + 2139000y + 18000s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 1622*y^8 + 4935721*y^6 - 855811400*y^4 - 4833806760000*y^2 +
8468214984000000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{12837823}{2638790623394114} + (-\frac{265}{10555162493576456})\,s, \qquad J = \begin{pmatrix} 2114 & 1252 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2114$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3778486223035968 = 2^6 \cdot 3^2 \cdot 17^2 \cdot 23^2 \cdot 71 \cdot 604343$:

$$\begin{pmatrix} 201325992276 & 48437162892 & 186550725962 & 173437118890 & 46967513682 & 187115361690 & 110212456565 & 131371418730 & 156492496497 & 2727625776 \\ 0 & 2346 & 920 & 1470 & 1656 & 2300 & 1478 & 1968 & 1096 & 1651 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 257 factors with signed exponents of absolute value up to 739988415; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 811y^3 - 10sy^2 + 2139000y + 18000s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 1622*y^8 + 4935721*y^6 - 855811400*y^4 - 4833806760000*y^2 +
8468214984000000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{5708627}{537850646109698} + (\frac{793}{2151402584438792})\,s, \qquad J = \begin{pmatrix} 1538 & 1082 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1538$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $132544212316560 = 2^4 \cdot 3^2 \cdot 5 \cdot 11^2 \cdot 113 \cdot 13463701$:

$$\begin{pmatrix} 502061410290 & 104507961162 & 164171693970 & 151846119472 & 411528651695 & 80089077375 & 226205633178 & 21702133564 & 310306319760 & 73779729730 \\ 0 & 22 & 10 & 4 & 6 & 1 & 16 & 20 & 18 & 11 \\ 0 & 0 & 6 & 0 & 2 & 5 & 0 & 4 & 2 & 3 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 257 factors with signed exponents of absolute value up to 495252930; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 1 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (s - 1207)\, y^3 + (19s - 62198)\, y^2 + (80s - 499216)\, y + (2696s + 2528496)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2414*y^8 - 124396*y^7 + 26594883*y^6 + 1148388672*y^5 + 18690
797414*y^4 + 276379365504*y^3 + 2779583877408*y^2 + 8749694655488*y
+ 196363999679872
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-2 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad -1 \quad 1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{144237523}{20805604708146875} + \left(\frac{209}{20805604708146875}\right) s, \qquad J = \begin{pmatrix} 1835 & 997 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1835$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $151988111174592 = 2^6 \cdot 3^2 \cdot 7 \cdot 11^2 \cdot 29 \cdot 2539 \cdot 4231$:

$$\begin{pmatrix} 575712542328 & 404844853704 & 255684106696 & 456043302156 & 95109025090 & 27745198240 & 7446681596 & 357758666293 & 489047208417 & 5428937199 \\ 0 & 6 & 4 & 3 & 0 & 3 & 2 & 0 & 5 & 3 \\ 0 & 0 & 44 & 1 & 34 & 42 & 0 & 5 & 39 & 43 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 263 factors with signed exponents of absolute value up to 28494769746; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (s - 1207)\, y^3 + (19s - 62198)\, y^2 + (80s - 499216)\, y + (2696s + 2528496)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2414*y^8 - 124396*y^7 + 26594883*y^6 + 1148388672*y^5 + 18690
797414*y^4 + 276379365504*y^3 + 2779583877408*y^2 + 8749694655488*y
+ 196363999679872
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-2 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad -1 \quad 1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{12837823}{2638790623394114} + \left(-\frac{265}{10555162493576456}\right) s, \qquad J = \begin{pmatrix} 2114 & 1252 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $40780756169216 = 2^9 \cdot 7^2 \cdot 37 \cdot 43932661$:

$$\begin{pmatrix} 45514236796 & 27366352076 & 37100911420 & 28887572238 & 13077434332 & 10886100648 & 44478105468 & 36457426951 & 43014106695 & 23094491949 \\ 0 & 28 & 14 & 26 & 26 & 16 & 8 & 27 & 7 & 9 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 268 factors with signed exponents of absolute value up to 31362128195; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (s - 1207)\, y^3 + (19s - 62198)\, y^2 + (80s - 499216)\, y + (2696s + 2528496)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2414*y^8 - 124396*y^7 + 26594883*y^6 + 1148388672*y^5 + 18690
797414*y^4 + 276379365504*y^3 + 2779583877408*y^2 + 8749694655488*y
+ 196363999679872
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-2 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad -1 \quad 1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{5708627}{537850646109698} + \left(\frac{793}{2151402584438792}\right) s, \qquad J = \begin{pmatrix} 1538 & 1082 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1538$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $179749363531248 = 2^4 \cdot 3^2 \cdot 7 \cdot 859 \cdot 207593459$:

$$\begin{pmatrix}
7489556813802 & 584529739692 & 3385065733132 & 4187639608936 & 1334474812364 & 3397202460403 & 5083613412401 & 2425458178383 & 4961181185592 & 3341679638102 \\
0 & 6 & 0 & 2 & 1 & 5 & 1 & 1 & 4 & 2 \\
0 & 0 & 2 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 2 & 1 & 1 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 266 factors with signed exponents of absolute value up to 53441324936; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.